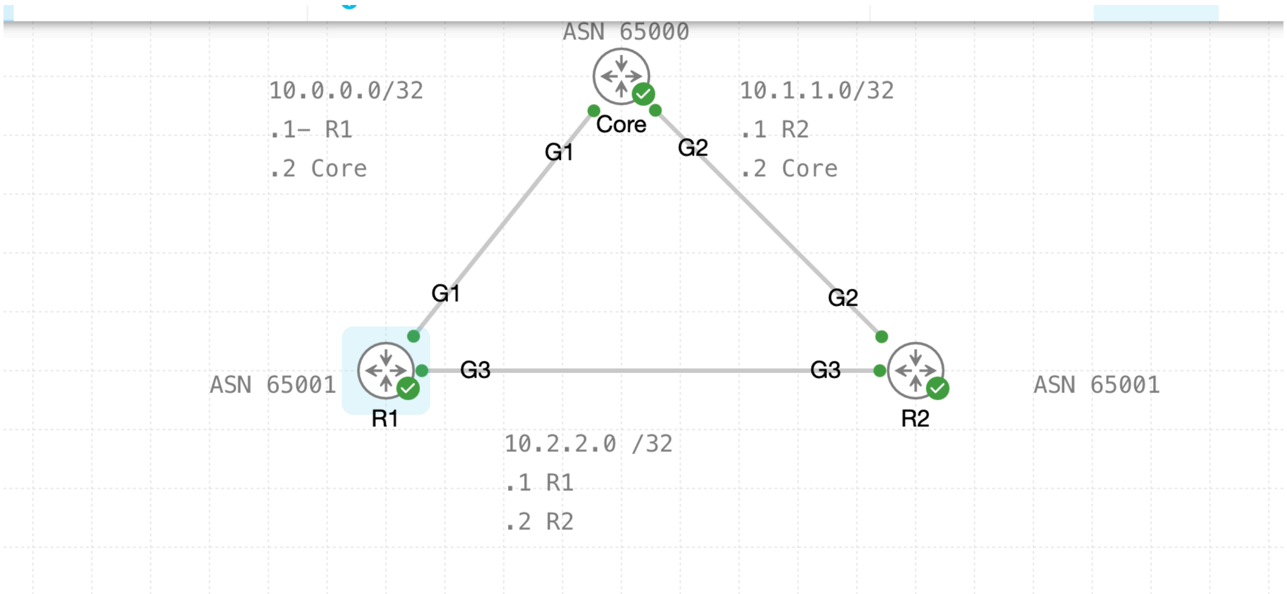
BGP- Local Preference

Friday, August 21, 2026 12:14 PM

Topology



R1 before Local preference. The path is taking 10.0.0.2 Core's interface.

```
Loose, Strict, Record, Timestamp, Verbose[none]:
Type escape sequence to abort.
Tracing the route to 100.100.100.2
VRF info: (vrf in name/id, vrf out name/id)
  1 10.0.0.2 1 msec 1 msec *
R1(config-router-af)#do show ip bgp
BGP table version is 23, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
```

Local preference change : Prefix list and route-map created

```
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#show ip prefix-lis
                    ^
% Invalid input detected at '^' marker.

R1(config)#do show ip prefix-lis
ip prefix-list LP1: 1 entries
   seq 5 permit 1.1.1.1/32
R1(config)#do show route-ma
route-map set-local-p, permit, sequence 10
```

```
  Set clauses:
  Policy routing matches: 0 packets, 0 bytes
R1(config)#
```

R1- BGP settings .

```
R1(config-router-af)#do show run | sec router bgp
router bgp 65001
 bgp log-neighbor-changes
 neighbor 10.0.0.2 remote-as 65000
 neighbor 10.2.2.2 remote-as 65001
 !
 address-family ipv4
  network 1.1.1.1 mask 255.255.255.255
  network 10.0.0.0 mask 255.255.255.252
  network 10.2.2.0 mask 255.255.255.252
  neighbor 10.0.0.2 activate
  neighbor 10.0.0.2 as-override
  neighbor 10.0.0.2 allowas-in
  neighbor 10.2.2.2 activate
  neighbor 10.2.2.2 route-map set-local-p out
 exit-address-family
R1(config-router-af)#do show run | sec route-map
  neighbor 10.2.2.2 route-map set-local-p out
route-map set-local-p permit 10
 match ip address prefix-list LP1
 set local-preference 500
route-map set-local-p permit 20
R1(config-router-af)#
```

R1 for 1.1.1.1 LP 500

```
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   1.1.1.1/32       0.0.0.0                  0         32768 i
 *>   2.2.2.2/32       10.0.0.2                      500      0 65000 65001 i
 *>   10.0.0.0/30      0.0.0.0                  0         32768 i
 *>   10.1.1.0/30      10.0.0.2                      500      0 65000 65001 i
 *>   100.100.100.2/32 10.0.0.2                 0    500      0 65000 i
R1(config-router-af)#
```

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R2: shows routes set for IP Prefix with a LP of 250

```
R2#show run | sec route-map
  neighbor 10.2.2.1 route-map set-local-p out
route-map set-local-p permit 10
 match ip address LP2
 set local-preference 250
```

```
  network 10.1.1.0 mask 255.255.255.252
  network 10.2.2.0 mask 255.255.255.252
  neighbor 10.1.1.2 activate
  neighbor 10.1.1.2 as-override
  neighbor 10.1.1.2 allowas-in
  neighbor 10.2.2.1 activate
  neighbor 10.2.2.1 route-map set-local-p out
 exit-address-family
R2#
```

****

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BGP settings for all routers in the network topology

R1 : Interface and BGP settings

```
R1(config)#
*Aug 21 16:17:39.165: %SMART_LIC-6-HOSTNAME_MATCHED_UDI: The host name has been changed to match a field in the de
 the device identifier is sent to Cisco this may bypass your host name privacy settingsdo show ip int br | sec up
GigabitEthernet1       10.0.0.1        YES manual up                    up
Loopback1              1.1.1.1         YES manual up                    up
R1(config)#do show run | sec router bgp
router bgp 65001
 bgp log-neighbor-changes
 neighbor 10.0.0.2 remote-as 65000
 !
 address-family ipv4
  network 1.1.1.1 mask 255.255.255.255
  network 10.0.0.0 mask 255.255.255.252
  neighbor 10.0.0.2 activate
  neighbor 10.0.0.2 as-override
  neighbor 10.0.0.2 allowas-in
 exit-address-family
R1(config)#
```

HUB

```
Core(config)#do show run | sec router bgp
router bgp 65000
 bgp log-neighbor-changes
 neighbor 10.0.0.1 remote-as 65001
 neighbor 10.1.1.1 remote-as 65001
 !
 address-family ipv4
  network 100.100.100.2 mask 255.255.255.255
  neighbor 10.0.0.1 activate
  neighbor 10.1.1.1 activate
 exit-address-family
Core(config)#
```

R2:

```
R2(config)#do show ip int br | sec up
GigabitEthernet2       10.1.1.1        YES manual up                    up
Loopback2              2.2.2.2         YES manual up                    up
R2(config)#do show run | sec router bgp
router bgp 65001
```

****

BGP table on R1 - Before Local Preference

```
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   1.1.1.1/32       0.0.0.0                  0         32768 i
 *>   2.2.2.2/32       10.0.0.2                               0 65000 65001 i
 *>   10.0.0.0/30      0.0.0.0                  0         32768 i
 *>   10.1.1.0/30      10.0.0.2                               0 65000 65001 i
 *>   100.100.100.2/32 10.0.0.2                 0             0 65000 i
```

BGP TABLE R2

```
BGP table version is 17, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   1.1.1.1/32       10.1.1.2                               0 65000 65001 i
 *>   2.2.2.2/32       0.0.0.0                  0         32768 i
 *>   10.0.0.0/30      10.1.1.2                               0 65000 65001 i
 *>   10.1.1.0/30      0.0.0.0                  0         32768 i
 *>   100.100.100.2/32 10.1.1.2                 0             0 65000 i
R2(config)#
```

****

After route-map applied- LP applied

R2 receiving LP 500- After R1 configured LP

```
R2(config-router-af)#do show ip bgp
BGP table version is 25, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>i  1.1.1.1/32       10.2.2.1                 0    500      0 i
 *                     10.1.1.2                               0 65000 65001 i
 *>   2.2.2.2/32       0.0.0.0                  0         32768 i
 *>i  10.0.0.0/30      10.2.2.1                 0    100      0 i
 *                     10.1.1.2                               0 65000 65001 i
 *>   10.1.1.0/30      0.0.0.0                  0         32768 i
 *>   10.2.2.0/30      0.0.0.0                  0         32768 i
 * i                   10.2.2.1                 0    100      0 i
 *                     10.1.1.2                               0 65000 65001 i
 * i  100.100.100.2/32 10.0.0.2                 0    100      0 65000 i
```

```
R1(config-router-af)#do show ip bgp
BGP table version is 20, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   1.1.1.1/32       0.0.0.0                  0         32768 i
 *>i  2.2.2.2/32       10.2.2.2                 0    250      0 i
 *                     10.0.0.2                               0 65000 65001 i
 *>   10.0.0.0/30      0.0.0.0                  0         32768 i
 *>i  10.1.1.0/30      10.2.2.2                 0    250      0 i
 *                     10.0.0.2                               0 65000 65001 i
 * i  10.2.2.0/30      10.2.2.2                 0    250      0 i
 *>                    0.0.0.0                  0         32768 i
 *>i  100.100.100.2/32 10.1.1.2                 0    250      0 65000 i
 *                     10.0.0.2                 0             0 65000 i
R1(config-router-af)#
```

!!
!!

Validation: R1

-Traceroute from R1 (1.1.1.1- source)

to the loopback address of Core

on 100.100.100.2. Traffic is going to R2 then upstream.

```
>_ R1    >_ CORE    >_ R2
Numeric display [n]:
Timeout in seconds [3]:
Probe count [3]:
Minimum Time to Live [1]:
Maximum Time to Live [30]: 5
Port Number [33434]:
Loose, Strict, Record, Timestamp, Verbose[none]:
Type escape sequence to abort.
Tracing the route to 100.100.100.2
VRF info: (vrf in name/id, vrf out name/id)
  1 10.2.2.2 1 msec 1 msec 1 msec
  2 10.1.1.2 44 msec 2 msec *
R1#
```

!!
!!

BGP MED

Core:

```
Core#show run | sec router bgp
router bgp 65000
 bgp log-neighbor-changes
 neighbor 10.0.0.1 remote-as 65001
 neighbor 10.1.1.1 remote-as 65001
 !
 address-family ipv4
  network 100.100.100.2 mask 255.255.255.255
  neighbor 10.0.0.1 activate
  neighbor 10.1.1.1 activate
 exit-address-family
Core#
```

CPU 12.58% MEMC

Core :BGP Table

```
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   1.1.1.1/32       10.1.1.1                               0 65001 i
 *                     10.0.0.1                 251           0 65001 i
 *    2.2.2.2/32       10.0.0.1                 251           0 65001 i
 *>                    10.1.1.1                 250           0 65001 i
 r>   10.0.0.0/30      10.1.1.1                               0 65001 i
 r                     10.0.0.1                 251           0 65001 i
 r    10.1.1.0/30      10.0.0.1                 251           0 65001 i
 r>                    10.1.1.1                   0           0 65001 i
 *>   10.2.2.0/30      10.1.1.1                   0           0 65001 i
 *                     10.0.0.1                 251           0 65001 i
 *    100.100.100.2/32 10.0.0.1                 251           0 65001 65001 i
 *>                    0.0.0.0                    0       32768 i
Core#
```

```
Core#show ip bgp 1.1.1.1
BGP routing table entry for 1.1.1.1/32, version 26
Paths: (2 available, best #1, table default)
  Advertised to update-groups:
     4
  Refresh Epoch 1
  65001
    10.1.1.1 from 10.1.1.1 (2.2.2.2)
      Origin IGP, localpref 100, valid, external, best
      rx pathid: 0, tx pathid: 0x0
      Updated on Aug 21 2026 21:02:06 UTC
  Refresh Epoch 2
  65001
    10.0.0.1 from 10.0.0.1 (1.1.1.1)
      Origin IGP, metric 251, localpref 100, valid, external
      rx pathid: 0, tx pathid: 0
      Updated on Aug 21 2026 20:43:13 UTC
Core#
```

```
    ip address prefix-lists: med
  Set clauses:
    metric 251
  Policy routing matches: 0 packets, 0 bytes
route-map med1, permit, sequence 20
  Match clauses:
  Set clauses:
  Policy routing matches: 0 packets, 0 bytes
```

R2 route map

```
R2>en
R2#show route-map | sec med
route-map med, permit, sequence 10
  Match clauses:
    ip address prefix-lists: LP2
  Set clauses:
    metric 250
  Policy routing matches: 0 packets, 0 bytes
route-map med, permit, sequence 20
  Match clauses:
  Set clauses:
  Policy routing matches: 0 packets, 0 bytes
```

VALIDATION:

Core to loop1 on R1 goes to R2

```
Loose, Strict, Record, Timestamp, Verbose[none]:
Type escape sequence to abort.
Tracing the route to 1.1.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 10.1.1.1 [AS 65001] 1 msec 1 msec 1 msec
  2 10.2.2.1 [AS 65001] 1 msec 1 msec *
Core#show ip int br | sec up
```

R2 int gi2

```
R2#show ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet1       unassigned      YES unset  administratively down down
GigabitEthernet2       10.1.1.1        YES manual up                    up
GigabitEthernet3       10.2.2.2        YES manual up                    up
GigabitEthernet4       unassigned      YES unset  administratively down down
Loopback2              2.2.2.2         YES manual up                    up
R2#
```